BASE CHANGE ISOMORPHISM FOR COMPACTLY SUPPORTED DERIVED PUSHFORWARD ALONG SEPARATED FINITE TYPE MORPHISMS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-etale.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Editorial note. Disclosure: the assembly argument is omitted verbatim – 'A straightforward argument which we omit shows that both follow if we can show that the isomorphisms (1), (2), (3) are compatible with base change' (proof lines 3690-3691). What is omitted is the bookkeeping assembly from the three generating compatibilities to the two deliverables (independence of the compactification and stacking compatibility); the conclusion is stated precisely and every input is anchored: (1) by the Sites composing-base-change remark, (2) by the chapter remark on base change compatibility of the composition isomorphism, and (3) by lemma-shriek-proper-and-open-base-change. The compactification-independence part parallels the cofiltered argument spelled out in the proof of lemma-shriek-well-defined. No other step is omitted.

Allowed standard prerequisites (editorial).

- Proper base change (Etale Cohomology, Lemmas etale-cohomology-lemma-proper-base-change 0DDE and etale-cohomology-lemma-proper-base-change-mod-n 0F0C). cited at proof lines 3675-3680. The first arrow of the construction: base change for the compactification pushforward, in the bounded below torsion and torsion ring cases respectively.
- Base change for separated lower shriek (Lemma lemma-base-change-f-shriek-separated, canonical tag 0F55). cited at proof lines 3680-3681. The second arrow of the construction: the isomorphism for the lower shriek part, valid before sheafification issues since it is the abelian-sheaf level iso.
- Base change compatibility of the exchange map (Lemma lemma-shriek-proper-and-open-base-change, canonical tag 0F7F). cited at proof line 3704. Anchors compatibility (3), the exchange-map input of the assembly argument.

Source reference. Original lemma 0DDE: sources/stacks-chow/tags.txt, source lines 16764–16764. Cited at proof lines 3675-3677: the first arrow of the construction in the bounded below torsion case.

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Source reference. Original lemma 0F0C: sources/stacks-chow/tags.txt, source lines 18757–18757. Cited at proof lines 3678-3680: the first arrow of the construction when Lambda is a torsion ring.

Source reference. Original lemma 0F55: sources/stacks-chow/tags.txt, source lines 18925–18925. Cited at proof lines 3680-3681: the second arrow of the construction.

Source reference. Original lemma 0F7J: sources/stacks-chow/tags.txt, source lines 19009–19009. Cited at statement lines 3659-3660 and proof line 3689: the composition isomorphisms with which the base change isomorphisms must be compatible.

Source reference. Original lemma 0F7F: sources/stacks-chow/tags.txt, source lines 19005–19005. Cited at proof line 3704: anchors compatibility (3) of the exchange map with base change.

Source reference. Original reference 0F77: sources/stacks-chow/tags.txt, source lines 18997–18997. Cited at proof line 3703: anchors compatibility (2) of the lower shriek composition isomorphism with base change.

Source reference. Original reference 0E46: sources/stacks-chow/tags.txt, source lines 17666–17666. Cited at proof lines 3701-3702: anchors compatibility (1) of the composition of proper pushforwards with base change.

Lemma 0.2. *Consider a cartesian square*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

of quasi-compact and quasi-separated schemes with f separated and of finite type. Then there is a canonical isomorphism

$$g^{-1} \circ Rf_! \to Rf'_! \circ (g')^{-1}$$

Moreover, these isomorphisms are compatible with the isomorphisms of Lemma 0F7J.

Proof. Choose a compactification $j : X \to \overline{X}$ over Y and denote $\overline{f} : \overline{X} \to Y$ the structure morphism. Let $j' : X' \to \overline{X}'$ and $\overline{f}' : \overline{X}' \to Y'$ denote the base changes of j and $\overline{f}$. Since $Rf_! = R\overline{f}_* \circ j_!$ and $Rf'_! = R\overline{f}'_* \circ j'_!$ the isomorphism can be constructed via

$$g^{-1} \circ R\overline{f}_* \circ j_! \to R\overline{f}'_* \circ (\overline{g}')^{-1} \circ j_! \to R\overline{f}'_* \circ j'_! \circ (g')^{-1}$$

where the first arrow is the isomorphism given to us by the proper base change theorem (Étale Cohomology, Lemma 0DDE in the bounded below torsion case and Étale Cohomology, Lemma 0F0C in the case that Λ is torsion) and the second arrow is the isomorphism of Lemma 0F55.

To finish the proof we have to show two things: first we have to show that the isomorphism of functors so obtained does not depend on the choice of the compactification and second we have to show that if we vertically stack two base change diagrams as in the lemma, then these base change isomorphisms are compatible with the isomorphisms of Lemma 0F7J. A straightforward argument which we omit shows that both follow if we can show that the isomorphisms

(1) $Rg_* \circ Rf_* = R(g \circ f)_*$ for $f : X \to Y$ and $g : Y \to Z$ proper,
(2) $g_! \circ f_! = (g \circ f)_!$ for $f : X \to Y$ and $g : Y \to Z$ separated and quasi-finite, and
(3) $g_! \circ Rf'_* = Rf_* \circ g'_!$ for $f : X \to Y$ and $f' : X' \to Y'$ proper and $g : Y' \to Y$ and $g' : X' \to X$ separated and quasi-finite with $f \circ g' = g \circ f'$

are compatible with base change. This holds for (1) by Cohomology on Sites, Remark 0E46, for (2) by Remark 0F77, and (3) by Lemma 0F7F. □